

Module 1

DC Circuit Analysis and Theorems

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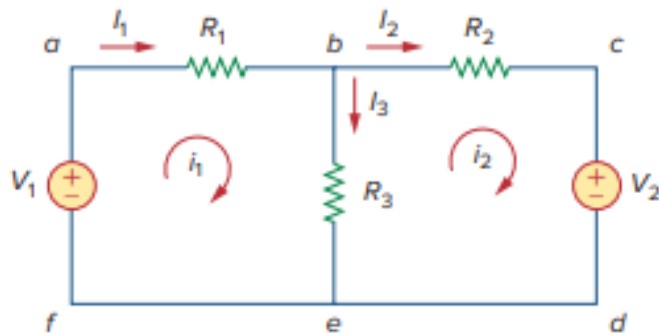
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Mesh analysis

Mesh analysis applies KVL to find unknown currents.

Steps to determine Node Voltages:

1. Assign mesh currents $i_1, i_2, \dots$ in to the n meshes.
2. Apply KVL to each of the n meshes.
3. Use Ohm's law to express the voltages in terms of the mesh currents.
4. Solve the resulting n simultaneous equations to get the mesh currents.



Mesh 1,

$$-V_1 + R_1 i_1 + R_3 (i_1 - i_2) = 0$$

$$(R_1 + R_3) i_1 - R_3 i_2 = V_1 \quad \text{-----1}$$

Mesh 2,

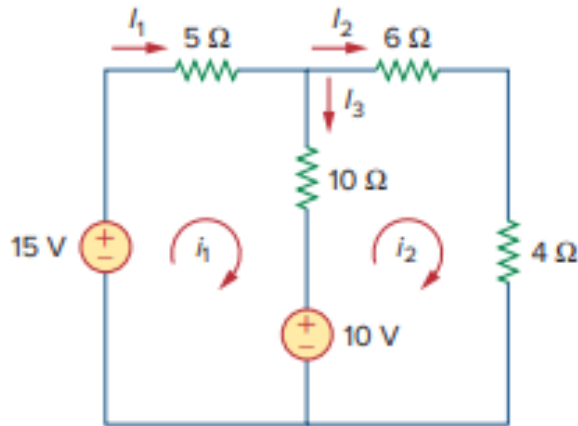
$$R_2 i_2 + V_2 + R_3 (i_2 - i_1) = 0$$

$$-R_3 i_1 + (R_2 + R_3) i_2 = -V_2 \quad \text{-----2}$$

Solve the 2 simultaneous equations to get i_1 and i_2 .

Mesh analysis

In the circuit below, find the branch currents I_1 , I_2 , and I_3 using mesh analysis.



Mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

Mesh 2,

$$3i_1 - 2i_2 = 1 \quad \text{-----1}$$

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

$$i_1 = 2i_2 - 1 \quad \text{-----2}$$

Solve the 2 simultaneous equations:

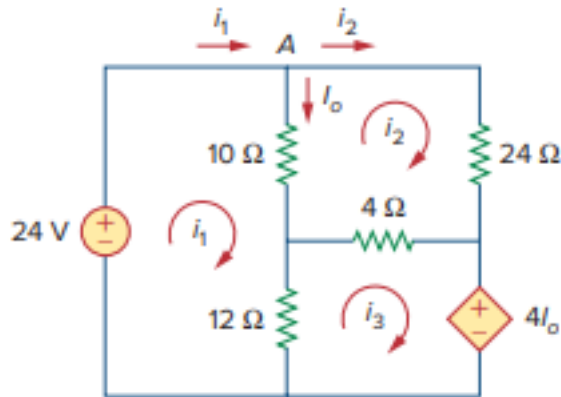
$$I_1 = 1\text{A}$$

$$I_2 = 1\text{A}$$

Hence $I_3 = 0$

Mesh analysis

Use mesh analysis to find the current I_o in the circuit .



Mesh 3,

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

At node A, $i_1 - i_2 = I_o$

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

Mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

$$-i_1 - i_2 + 2i_3 = 0 \quad \text{-----}3$$

$$11i_1 - 5i_2 - 6i_3 = 12 \quad \text{-----}1$$

Mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

$$-5i_1 + 19i_2 - 2i_3 = 0 \quad \text{-----}2$$

Solve the 3 simultaneous equations:

$$i_1 = 2.25\text{A}$$

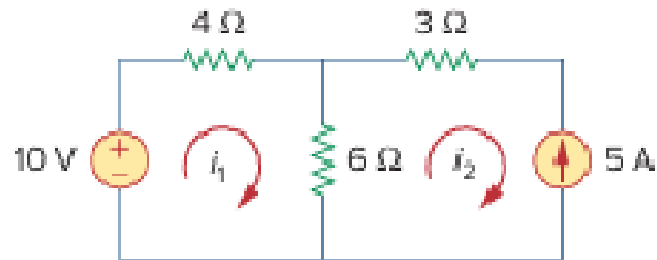
$$i_2 = 0.75\text{A}$$

$$I_3 = 1.5\text{ A}$$

Hence $I_o = 1.5\text{ A}$

Mesh analysis

Find the loop currents.



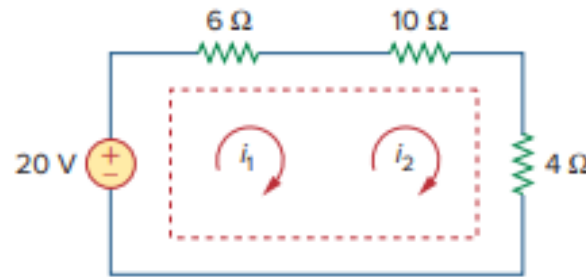
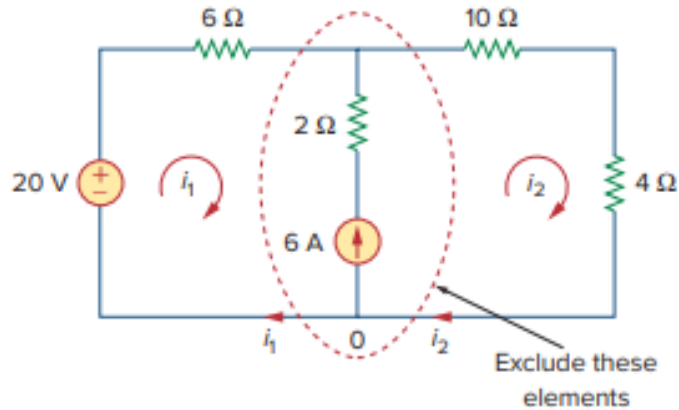
$$i_2 = -5\text{ A}$$

$$-10 + 4i_1 + 6(i_1 - i_2) = 0 \quad \Rightarrow \quad i_1 = -2\text{ A}$$

Supermesh

A supermesh results when two meshes have a (dependent or independent) **current source in common**.

A supermesh requires the application of both KVL and KCL.



We create a supermesh and apply KVL to the supermesh.

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0 \quad \text{or} \quad 6i_1 + 14i_2 = 20 \quad \text{-----}1$$

Apply KCL to a node in the branch where the two meshes intersect. Applying KCL to node 0,

$$i_2 = i_1 + 6 \quad \text{-----}2$$

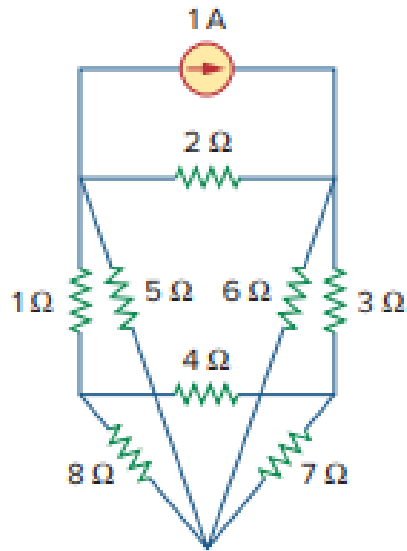
Solving equations 1 & 2 we get, $i_1 = -3.2 \text{ A}$, $i_2 = 2.8 \text{ A}$

Mesh analysis

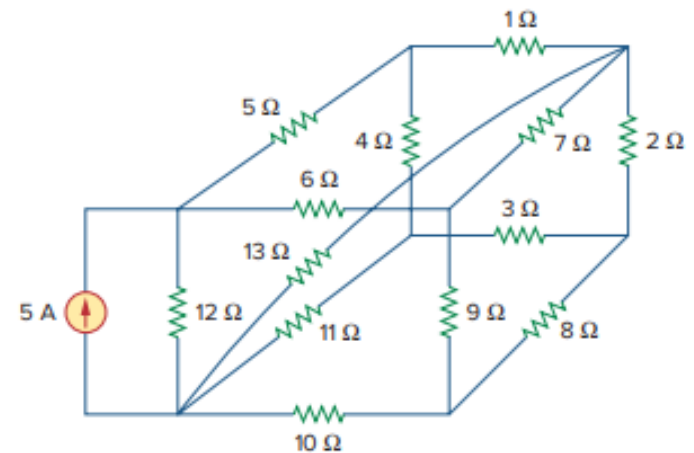
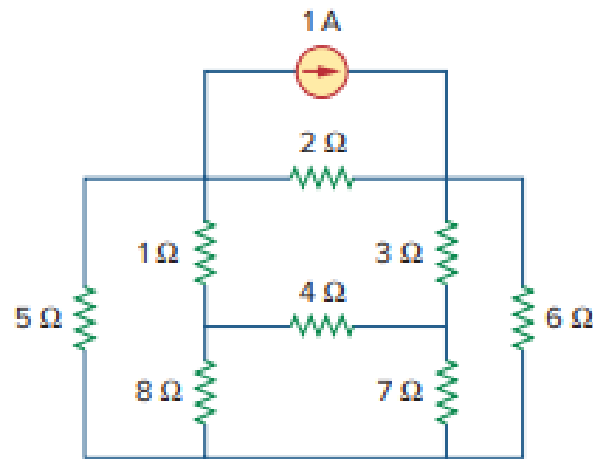
Mesh analysis is applicable to a circuit that is planar.

A planar circuit is one that can be drawn in a plane with no branches crossing one another; otherwise it is non-planar.

A circuit may have crossing branches and still be planar if it can be redrawn such that it has no crossing branches.



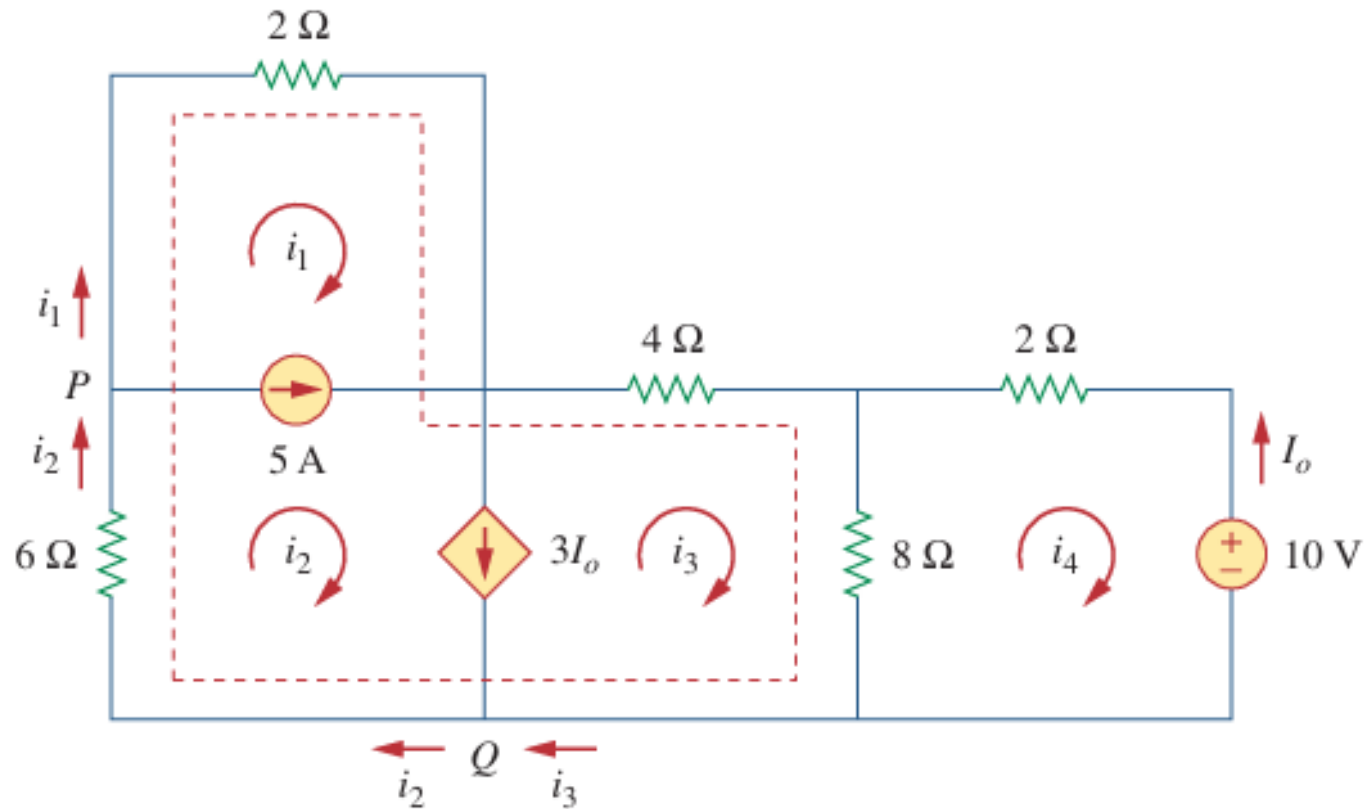
planar



non-planar

Supermesh

For the circuit below, calculate i_1 , i_2 , i_3 , i_4 .



Supermesh

Meshes 1 and 2 form a supermesh since they have an independent current source in common.

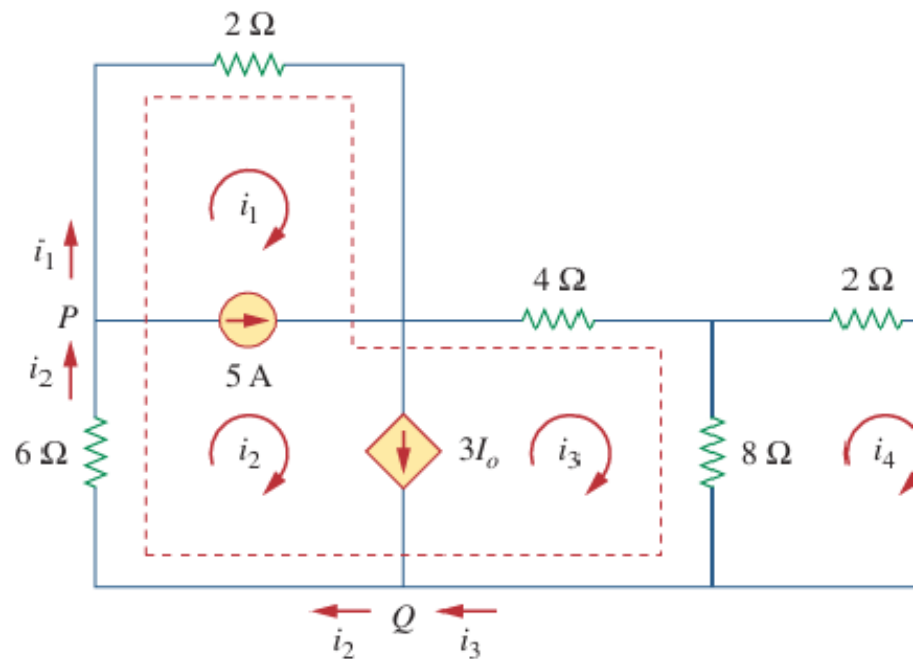
Also, meshes 2 and 3 form another supermesh because they have a dependent current source in common.

The two supermeshes intersect and form a larger supermesh.

Apply KVL to the larger supermesh,

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

$$i_1 + 3i_2 + 6i_3 - 4i_4 = 0 \quad \text{-----} 1$$



Supermesh

For the independent current source, we apply KCL to node P:

$$i_2 = i_1 + 5 \quad \text{-----}2$$

For the dependent current source, we apply KCL to node Q:

$$i_2 = i_3 + 3I_o$$

But $I_o = -i_4$, hence,

$$i_2 = i_3 - 3i_4 \quad \text{-----}3$$

Applying KVL in mesh 4,

$$2i_4 + 8(i_4 - i_3) + 10 = 0$$

$$5i_4 - 4i_3 = -5 \quad \text{-----}4$$

Solving the eqns 1, 2, 3, 4 we find

$$i_1 = -7.5 \text{ A}, \quad i_2 = -2.5 \text{ A}, \quad i_3 = 3.93 \text{ A}, \quad i_4 = 2.143 \text{ A}$$

Next class: Superposition theorem